Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 1 of 17 · Question 6 part a.iv · 3 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part a.iv in this mark scheme extract

| | | | |
|---|---|---|---|
| 6 (a)(i) | A aorta; | | 1 |
| (ii) | D vena cava; | | 1 |
| (iii) | • wider lumen;<br>• thinner walls;<br>• valves;<br>  | Ignore arrows<br>Accept valves drawn either way round | 1<br>1<br>1 |
| (iv) | Any three from:<br><br>(Blood vessel X)<br>• thicker walls/more muscle/elastic fibres in walls to maintain blood flow to maintain/withstand high (blood) pressure;<br>• thinner lumen to maintain high blood pressure;<br><br>OR<br><br>(Blood vessel Y) -<br>• less muscle/elastic tissue in walls so slower blood flow;<br>• wider lumen so carries blood at lower pressure;<br><br>• contains valves to prevent backflow; | | Max 3 |

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 1 of 17 · Question 6 part a.iv · 3 marks

2019 JUN · Paper 02 · extract 2 of 2 · find part a.iv in this mark scheme extract

| | | | |
|---|---|---|---|
| (b) | Any four from:<br>• thin wall/wall one cell thick;<br>• contains pores;<br>• shorter diffusion pathway;<br>• narrow lumen;<br>• only one (blood) cell/few (blood) cells at a time can pass through; | | Max 4 |

**Total question 6 = 12 marks**

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 2 of 17 · Question 6 part a.iii · 3 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part a.iii in this mark scheme extract

| | | | |
|---|---|---|---|
| 6 (a)(i) | A aorta; | | 1 |
| (ii) | D vena cava; | | 1 |
| (iii) | • wider lumen;<br>• thinner walls;<br>• valves;<br>  | Ignore arrows<br>Accept valves drawn either way round | 1<br>1<br>1 |
| (iv) | Any three from:<br><br>(Blood vessel X)<br>• thicker walls/more muscle/elastic fibres in walls to maintain blood flow to maintain/withstand high (blood) pressure;<br>• thinner lumen to maintain high blood pressure;<br><br>OR<br><br>(Blood vessel Y) -<br>• less muscle/elastic tissue in walls so slower blood flow;<br>• wider lumen so carries blood at lower pressure;<br><br>• contains valves to prevent backflow; | | Max 3 |

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 2 of 17 · Question 6 part a.iii · 3 marks

2019 JUN · Paper 02 · extract 2 of 2 · find part a.iii in this mark scheme extract

| | | | |
|---|---|---|---|
| (b) | Any four from:<br>• thin wall/wall one cell thick;<br>• contains pores;<br>• shorter diffusion pathway;<br>• narrow lumen;<br>• only one (blood) cell/few (blood) cells at a time can pass through; | | Max 4 |

**Total question 6 = 12 marks**

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 3 of 17 · Question 1 part c · 3 marks

2022 JUN · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

| | | | |
|---|---|---|---|
| 1 (a) | pulmonary artery; | | 1 |
| (b) | In the following order:<br>• oxygenated;<br>• lungs;<br>• atrium;<br>• ventricle;<br>• body; | | 5 |
| (c) | • blood vessel X has a thicker wall;<br>• blood vessel X has a smaller lumen;<br>• blood vessel X does not contain valves | Allow the reverse argument for blood vessel Y | 3 |
| | | | Total 9 |

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 3 of 17 · Question 1 part c · 3 marks

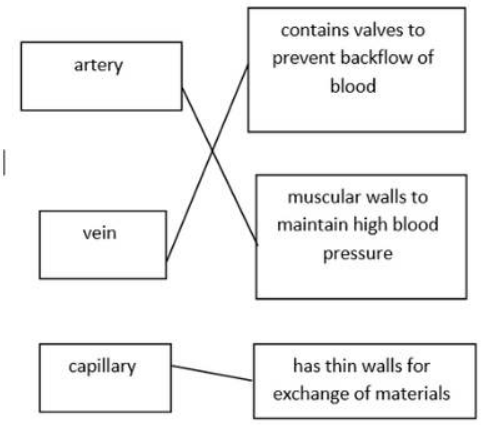
2022 JUN · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 4 of 17 · Question 2 part b · 2 marks

2024 JUN · Paper 01 (regional R) · find part b in this mark scheme extract

| | | | |
|---|---|---|---|
| 2 (a) (i) | X = plasma;<br>Y = red blood cell;<br>Z = white blood cell/phagocyte; | Accept named phagocyte | (3) |
| (a) (ii) | • transports oxygen;<br>• to body/ cells/tissues/organs; | | (2) |
| (a) (iii) | the patient has an infection/leukaemia/allergy/ blood cancer, inflammation; | Accept pathogens in body | (1) |
| (b) |  | Reject more than one line from each blood vessel<br>3 correct=2marks<br>1 or 2 correct= 1mark | (2) |
| (c) (i) | left ventricle; | | (1) |
| (c) (ii) | • into right ventricle;<br>• through pulmonary artery;<br>• to the lungs;<br>• (from the lungs) into the pulmonary vein; | | (4) |

**Total for Question 2 = 13 marks**

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 5 of 17 · Question 6 part b · 4 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

| | | | |
|---|---|---|---|
| 6 (a)(i) | A aorta; | | 1 |
| (ii) | D vena cava; | | 1 |
| (iii) | • wider lumen;<br>• thinner walls;<br>• valves;<br>  | Ignore arrows<br>Accept valves drawn either way round | 1<br>1<br>1 |
| (iv) | Any three from:<br><br>(Blood vessel X)<br>• thicker walls/more muscle/elastic fibres in walls to maintain blood flow to maintain/withstand high (blood) pressure;<br>• thinner lumen to maintain high blood pressure;<br><br>OR<br><br>(Blood vessel Y) -<br>• less muscle/elastic tissue in walls so slower blood flow;<br>• wider lumen so carries blood at lower pressure;<br><br>• contains valves to prevent backflow; | | Max 3 |

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 5 of 17 · Question 6 part b · 4 marks

2019 JUN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

| | | | |
|---|---|---|---|
| (b) | Any four from:<br><br>• thin wall/wall one cell thick;<br>• contains pores;<br>• shorter diffusion pathway;<br>• narrow lumen;<br>• only one (blood) cell/few (blood) cells at a time can pass through; | | Max 4 |

**Total question 6 = 12 marks**

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 6 of 17 · Question 4 part d · 2 marks

2020 JAN · Paper 01 · find part d in this mark scheme extract

| Question number | | | | |
|---|---|---|---|---|
| 4 | (a) | • C (left) atrium( 1)<br>• D (left) ventricle (1) | Maximum of one mark if right atrium/ventricle given | 2 |
| | (b) | Any four from the following:<br>• (blood vessels) A and F are arteries (1)<br>• A is the aorta/F is the pulmonary artery (1)<br>• arteries carry blood away from the heart (1)<br>• blood vessels B and E are veins (1)<br>• B is the vena cava/E is the pulmonary vein (1)<br>• veins carry blood towards the heart (1) | | 4 |
| | (c) | the heart pumps blood around the body/liver detoxifies the blood/involved in digestion / homeostasis/liver not muscular/liver has no chambers (1) | | 1 |
| | (d) | (capillaries) exchange of materials( between the blood and body tissues/cells)/allows diffusion (1) named substance (1) | | 2 |

**Total for Question 4 = 9 marks**

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 7 of 17 · Question 8 part a.ii · 2 marks

2020 JAN · Paper 01 · find part a.ii in this mark scheme extract

| Number | | | | |
|---|---|---|---|---|
| 8 | (a) (i) | X = urethra (1)<br>Y = ureter (1) | | 2 |
| | (ii) | Any two from:<br>• transports oxygenated blood from the heart (1)<br>• to the body/named organ (1)<br>• maintains high blood pressure (1) | | 2 |
| | (iii) | Any two from<br>• blood in A/renal artery contains more oxygen/oxygenated (1)<br>• blood in A/renal artery contains more urea (1)<br>• blood in A/renal artery contains more ions (1)<br>• similar concentration of glucose/B renal vein has less glucose (1)<br>• blood in A/renal artery contains less carbon dioxide (1) | Allow reverse argument for B/renal vein | 2 |
| | (b) (i) | Any four from:<br>• dialysis fluid/dialysate contains a low(er) concentration/no urea/waste products/ions (1)<br>• urea/waste/ions diffuse from blood (1)<br>• through dialysis membrane (1)<br>• from high(er) concentration (in the blood) to a low(er) concentration (in the dialysate)/down a concentration gradient (1)<br>• dialysis membrane is partially-permeable(1) | ORA<br><br>allow selectively permeable membrane | 4 |
| | (ii) | • to prevent blood from clotting whilst in the machine (1)<br>• (added as it enters the machine) to prevent clots from blocking the machine (as blood flows through) (1)<br>• (not added at the end) as blood clotting properties need to return to normal before passing back into patient (1) | | 3 |

**Total for question 8 = 13 marks**

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 8 of 17 · Question 5 part b · 4 marks

2025 JUN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

| | | | |
|---|---|---|---|
| 5 (a) | • ACE inhibitor prevents/inhibits action of enzyme;<br>• prevents formation of angiotensin II;<br>• less( angiotensin ii) bind with receptors;<br>• Losartan blocks/binds to receptors;<br>• angiotensin ii cannot bind to receptor; | | 4 |
| (b) | • arteries/arterioles;<br>• muscles in walls;<br>• can contract/constrict;<br>• narrow lumen;<br>• smaller volume/space( for blood flow); | | 4 |
| (c) | • in normal/healthy kidney protein not found in urine;<br>• protein is a big molecule;<br>• should not pass through the glomerulus/basement membrane;<br>• damage to glomerulus/basement membrane; | | 2 |
| (d) | • stronger/intense/eq; <u>contractions</u>;<br>• (which causes) <u>muscle</u> to get thicker; | mp2 must be due to mp1 | 2 |
| (e) | • detoxification occurs in the liver;<br>• can't break down Losartan if impaired/accumulation /build-up in body/bloodstream/liver;<br>• results in (further) liver damage/liver failure/liver diseae; | | 2<br>**Total =14** |

## Type MS · Identify arteries, veins and capillaries and relate their structure to function

### Answer 8 of 17 · Question 5 part b · 4 marks

2025 JUN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|
| | • A=rods/cones/retina; | | |

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 9 of 17 · Question 1 part a.iii · 2 marks

2025 JUN · Paper 02 (regional R) · find part a.iii in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 1 (a) | (i) | long (bone) | | 1 |
| | (ii) | <table><tr><th rowspan="2">Part</th><th colspan="3">Function</th></tr><tr><th>reduces friction</th><th>produces red blood cells</th><th>provides strength</th></tr><tr><td>compact bone</td><td></td><td></td><td>✓</td></tr><tr><td>spongy bone</td><td></td><td>✓</td><td></td></tr><tr><td>cartilage</td><td>✓</td><td></td><td></td></tr></table> | | 3 |
| | (iii) | • provides nutrients/named nutrient/oxygen;<br>• removes waste/named waste | | 2 |
| (b) | (i) | In the following order:<br>• tendons;<br>• contracts;<br>• relaxes;<br>• pulls; | | 4 |
| | (ii) | 0.5 x 125;<br>62.5 (J); | Accept 63 (J) | 2 |
| | | | **Total for Question 1 = 12 marks** | |

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 10 of 17 · Question 1 part b.i · 1 mark

2021 NOV · Paper 02 · extract 1 of 2 · find part b.i in this mark scheme extract

| | | | |
|---|---|---|---|
| 1 (a) (i) | • transport oxygen;<br>• from lungs;<br>• to (body) tissues/cells; | Allow organs | 3 |
| (ii) | • haemoglobin to carry/combine with oxygen;<br>• no nucleus, so more haemoglobin;<br>• biconcave shape to increase surface area;<br>• small/flexible (cell) membrane to pass through capillaries;<br>• thin membrane to reduce diffusion distance; | | 4 |
| (b) (i) | vein; | | 1 |
| (ii) | • valve present/large lumen/thin (muscular) walls;<br>• only found in veins/not found in arteries/other blood vessels; | | 2 |
| (iii) | • distance 13-15 mm;<br>• divide by 10 = 1.3-1.5; | Ecf: max 1 mark if mp1 incorrect but division by 10 of their value is correct | 2<br>Total 12 |

| Question number | Answer | Notes | Marks |
|---|---|---|---|

## Type MS · Identify arteries, veins and capillaries and relate their structure to function

### Answer 10 of 17 · Question 1 part b.i · 1 mark

2021 NOV · Paper 02 · extract 2 of 2 · find part b.i in this mark scheme extract

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 11 of 17 · Question 1 part b.ii · 2 marks

2021 NOV · Paper 02 · extract 1 of 2 · find part b.ii in this mark scheme extract

| | | | |
|---|---|---|---|
| 1 (a) (i) | • transport oxygen;<br>• from lungs;<br>• to (body) tissues/cells; | Allow organs | 3 |
| (ii) | • haemoglobin to carry/combine with oxygen;<br>• no nucleus, so more haemoglobin;<br>• biconcave shape to increase surface area;<br>• small/flexible (cell) membrane to pass through capillaries;<br>• thin membrane to reduce diffusion distance; | | 4 |
| (b) (i) | vein; | | 1 |
| (ii) | • valve present/large lumen/thin (muscular) walls;<br>• only found in veins/not found in arteries/other blood vessels; | | 2 |
| (iii) | • distance 13-15 mm;<br>• divide by 10 = 1.3-1.5; | Ecf: max 1 mark if mp1 incorrect but division by 10 of their value is correct | 2<br>Total 12 |

| Question number | Answer | Notes | Marks |
|---|---|---|---|

## Type MS · Identify arteries, veins and capillaries and relate their structure to function

### Answer 11 of 17 · Question 1 part b.ii · 2 marks

2021 NOV · Paper 02 · extract 2 of 2 · find part b.ii in this mark scheme extract

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 12 of 17 · Question 3 part a.ii · 4 marks

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part a.ii in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 3 (a) | (i) | arrow from top to bottom; | | 1 |
| | (ii) | • A is an artery, thicker muscle wall;<br>• to pump blood from heart/to body;<br>• B is a vein with a valve present;<br>• to control direction of blood flow/to prevent backflow; | If no other marks scored award one for correct identification of both vessels | 4 |
| | (iii) | capillary; | | 1 |
| (b) | (i) | • decrease in deaths;<br>• in both sexes/males and females;<br>• between 2005 - 2014/over whole range; | | 3 |
| | (ii) | • 60 000 000 ÷ 100 000 = 600;<br>• 600 × 640;<br>• = 384 000; | | 3 |

Total 12 marks

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 12 of 17 · Question 3 part a.ii · 4 marks

2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part a.ii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 13 of 17 · Question 6 part a.i · 4 marks

2024 NOV · Paper 02 · extract 1 of 2 · find part a.i in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 6 (a) | (i) | • 1 = artery/ aorta/arterioles/named artery;<br>• 2 = vein/ venules/vena cava;<br>• 3 = capillary;<br>• 4 = vein; | | 4 |
| | (ii) | • 1 = thick wall/muscular wall/elastic wall/fixed/rigid shape/small lumen;<br>• 2 = thin wall/little muscle/little elastic tissue/ irregular shape/large lumen;<br>• 3 = one cell thick/pores/gaps;<br>• 4 = valves; | | 4 |
| (b) | | • M = (left) ventricle/heart;<br>• N = arteries/arterioles/aorta/named artery;<br>• O = capillaries;<br>• P = veins/venules/vena cava; | | 4 |

**Total 12 marks**

## Type MS · Identify arteries, veins and capillaries and relate their structure to function

### Answer 13 of 17 · Question 6 part a.i · 4 marks

2024 NOV · Paper 02 · extract 2 of 2 · find part a.i in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|
| | | | |

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 14 of 17 · Question 6 part a.ii · 4 marks

2024 NOV · Paper 02 · extract 1 of 2 · find part a.ii in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 6 (a) | (i) | • 1 = artery/ aorta/arterioles/named artery;<br>• 2 = vein/ venules/vena cava;<br>• 3 = capillary;<br>• 4 = vein; | | 4 |
| | (ii) | • 1 = thick wall/muscular wall/elastic wall/fixed/rigid shape/small lumen;<br>• 2 = thin wall/little muscle/little elastic tissue/ irregular shape/large lumen;<br>• 3 = one cell thick/pores/gaps;<br>• 4 = valves; | | 4 |
| (b) | | • M = (left) ventricle/heart;<br>• N = arteries/arterioles/aorta/named artery;<br>• O = capillaries;<br>• P = veins/venules/vena cava; | | 4 |

**Total 12 marks**

## Type MS · Identify arteries, veins and capillaries and relate their structure to function

### Answer 14 of 17 · Question 6 part a.ii · 4 marks

2024 NOV · Paper 02 · extract 2 of 2 · find part a.ii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|
| | | | |

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 15 of 17 · Question 3 part b · 2 marks

2020 NOV · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

| | | | |
|---|---|---|---|
| 3 (a) (i) | C; (red blood cells) | | 1 |
| (ii) | any three from<br>• more carbon dioxide (1)<br>• less oxygen (1)<br>• moister (1)<br>• warmer (1) | | 3 |
| (iii) | • large surface area (1)<br>• thin (walled)/wall one cell thick (1)<br>• good blood supply (1)<br>• moist lining (1) | | 3 |
| (b) | • thin wall (1)<br>• narrow lumen (1) | Allow references to only one blood cell passing through | 2 |
| (c) | • breathing (out)/exhalation more difficult/not smooth/shortness of breath/faster breathing rate (1)<br>• as lungs would not squeeze out air/forced exhalation(1) | | 2 |

Total 11 marks

## Type MS · Identify arteries, veins and capillaries and relate their structure to function

### Answer 15 of 17 · Question 3 part b · 2 marks

2020 NOV · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|
| | | | |

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 16 of 17 · Question 3 part a.i · 1 mark

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part a.i in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 3 (a) | (i) | arrow from top to bottom; | | 1 |
| | (ii) | • A is an artery, thicker muscle wall;<br>• to pump blood from heart/to body;<br>• B is a vein with a valve present;<br>• to control direction of blood flow/to prevent backflow; | If no other marks scored award one for correct identification of both vessels | 4 |
| | (iii) | capillary; | | 1 |
| (b) | (i) | • decrease in deaths;<br>• in both sexes/males and females;<br>• between 2005 - 2014/over whole range; | | 3 |
| | (ii) | • 60 000 000 ÷ 100 000 = 600;<br>• 600 × 640;<br>• = 384 000; | | 3 |

Total 12 marks

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 16 of 17 · Question 3 part a.i · 1 mark

2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part a.i in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 17 of 17 · Question 3 part a.iii · 1 mark

2022 JUN · Paper 02 (regional R) · extract 1 of 2 · find part a.iii in this mark scheme extract

| | | | | |
|---|---|---|---|---|
| 3 (a) | (i) | arrow from top to bottom; | | 1 |
| | (ii) | • A is an artery, thicker muscle wall;<br>• to pump blood from heart/to body;<br>• B is a vein with a valve present;<br>• to control direction of blood flow/to prevent backflow; | If no other marks scored award one for correct identification of both vessels | 4 |
| | (iii) | capillary; | | 1 |
| (b) | (i) | • decrease in deaths;<br>• in both sexes/males and females;<br>• between 2005 - 2014/over whole range; | | 3 |
| | (ii) | • 60 000 000 ÷ 100 000 = 600;<br>• 600 × 640;<br>• = 384 000; | | 3 |

Total 12 marks

Type MS · Identify arteries, veins and capillaries and relate their structure to function

Answer 17 of 17 · Question 3 part a.iii · 1 mark

2022 JUN · Paper 02 (regional R) · extract 2 of 2 · find part a.iii in this mark scheme extract

| Question number | Answer | Notes | Marks |
|---|---|---|---|